Type MS · Show and explain why impulses travel one way across a synapse

Answer 1 of 8 · Question 6 part b.iii · 4 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

Number				
6	(a)	<table><tr><th>Nervous system</th><th>Hormonal system</th></tr><tr><td>nerve fibres/cells;</td><td>blood;</td></tr><tr><td>electrical/electrochemical;</td><td>chemical;</td></tr><tr><td>fast;</td><td>slow;</td></tr></table>	Allow nerves / neurones Reject hormone	3
	(b) (i)	arrow drawn left to right;		1
	(ii)	• neurotransmitter; • released from synapse/synaptic knobs; • diffuses across gap; • bind to receptors on next cell;		4
	(iii)	• neurotransmitter only released at one side of synapse; • can only diffuse in one direction; • from high to low concentration; • receptors only present on post synaptic membrane;		4

Total 12 marks

Type MS · Show and explain why impulses travel one way across a synapse

Answer 1 of 8 · Question 6 part b.iii · 4 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Show and explain why impulses travel one way across a synapse

Answer 2 of 8 · Question 8 part b · 1 mark

2021 JUN · Paper 01 · find part b in this mark scheme extract

number				
8	(a)	any three from • rapid response; • automatic; • to protect body; • from harmful effects/named example;		3
	(b)	correct position;		1
	(c)	any six from • impulse arrives at end of axon; • synapse; • causes release of neurotransmitter; • e.g. acetylcholine; • from vesicles; • diffuses across gap/cleft; • combines with/stimulates (post synaptic) receptors; • triggers impulse;		6

Total 10 marks

Type MS · Show and explain why impulses travel one way across a synapse

Answer 3 of 8 · Question 3 part b.ii · 1 mark

2025 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

3	(a)		• rapid/fast/eq. ; • involuntary/automatic action; • does not involve conscious thought; • protective/prevents (further) damage;	allow instant-mp1 ignore ref to brain ignore sub-conscious ignore unconscious	3
	(b)	(i)	• sensory; • relay/intermediate; • motor;	correct positions on diagram	3
		(ii)	correct direction from sensory to motor;	Mark correctly if there are more than 1 arrow, unle/wrongss they are in opposite direction	1
		(iii)	synapse/synaptic cleft;		1
					Total = 8

Type MS · Show and explain why impulses travel one way across a synapse

Answer 3 of 8 · Question 3 part b.ii · 1 mark

2025 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Show and explain why impulses travel one way across a synapse

Answer 4 of 8 · Question 8 part a.i · 1 mark

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
8	(a)	(i)	arrow pointing from top to bottom (1)		1
		(ii)	• provides energy/ATP (1) • for synthesis of acetylcholine/neurotransmitter (1) • movement of vesicles/exocytosis (1)		3
	(b)	(i)	• impulses can't pass to intercostal muscles/diaphragm (1) • unable to contract and move ribs/increase volume of thorax/chest (1)		2
		(ii)	• prevents breakdown of neurotransmitter/acetylcholine (1) • which can compete with curare for/binds to, receptor sites (1)		2

Total for this question = 8 marks

Type MS · Show and explain why impulses travel one way across a synapse

Answer 4 of 8 · Question 8 part a.i · 1 mark

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

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Type MS · Show and explain why impulses travel one way across a synapse

Answer 5 of 8 · Question 6 part b.i · 1 mark

2022 JAN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
6	(a)	<table><tr><th>Nervous system</th><th>Hormonal system</th></tr><tr><td>nerve fibres/cells;</td><td>blood;</td></tr><tr><td>electrical/electrochemical;</td><td>chemical;</td></tr><tr><td>fast;</td><td>slow;</td></tr></table>	Allow nerves / neurones Reject hormone	3
	(b) (i)	arrow drawn left to right;		1
	(ii)	• neurotransmitter; • released from synapse/synaptic knobs; • diffuses across gap; • bind to receptors on next cell;		4
	(iii)	• neurotransmitter only released at one side of synapse; • can only diffuse in one direction; • from high to low concentration; • receptors only present on post synaptic membrane;		4

Total 12 marks

Type MS · Show and explain why impulses travel one way across a synapse

Answer 5 of 8 · Question 6 part b.i · 1 mark

2022 JAN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Show and explain why impulses travel one way across a synapse

Answer 6 of 8 · Question 3 part a.ii · 1 mark

2024 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

3 (a)	(i)	Synapse/synaptic cleft;		1
	(ii)	arrow pointing from top to bottom;		1
	(iii)	any six from • vesicles; • contain neurotransmitter/ named neurotransmitter; • (vesicles)fuse/binds/attaches with pre-synaptic membrane/(vesicles) moves to pre-synaptic membrane; • exocytosis/release of neurotransmitter; • which diffuses (across the gap/synapse)/description of diffusion; • binds/attaches with receptors on post-synaptic membrane; • (initiates)new impulse;		6
	(iv)	• neurotransmitter only released from pre-synaptic membrane/ neurone A/first neurone/(vesicles) with neurotransmitter in pre-synaptic knob/neurone A/first neurone; • diffuses down a concentration gradient/description of concentration gradient; • receptors only on post-synaptic membrane/neurone B/ second neurone;		3
(b)		any two from • addiction; • stroke/heart problems/high blood pressure; • steal to fund habit; • anxiety; • depression; • hallucinations; • memory loss; • brain damage/ damage nerve cells; • paranoia; • overdose/death;		2

Total 13 marks

Type MS · Show and explain why impulses travel one way across a synapse

Answer 6 of 8 · Question 3 part a.ii · 1 mark

2024 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Show and explain why impulses travel one way across a synapse

Answer 7 of 8 · Question 3 part a.iv · 3 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part a.iv in this mark scheme extract

3 (a)	(i)	Synapse/synaptic cleft;		1
	(ii)	arrow pointing from top to bottom;		1
	(iii)	any six from • vesicles; • contain neurotransmitter/ named neurotransmitter; • (vesicles)fuse/binds/attaches with pre-synaptic membrane/(vesicles) moves to pre-synaptic membrane; • exocytosis/release of neurotransmitter; • which diffuses (across the gap/synapse)/description of diffusion; • binds/attaches with receptors on post-synaptic membrane; • (initiates)new impulse;		6
	(iv)	• neurotransmitter only released from pre-synaptic membrane/ neurone A/first neurone/(vesicles) with neurotransmitter in pre-synaptic knob/neurone A/first neurone; • diffuses down a concentration gradient/description of concentration gradient; • receptors only on post-synaptic membrane/neurone B/ second neurone;		3
(b)		any two from • addiction; • stroke/heart problems/high blood pressure; • steal to fund habit; • anxiety; • depression; • hallucinations; • memory loss; • brain damage/ damage nerve cells; • paranoia; • overdose/death;		2

Total 13 marks

Type MS · Show and explain why impulses travel one way across a synapse

Answer 7 of 8 · Question 3 part a.iv · 3 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Show and explain why impulses travel one way across a synapse

Answer 8 of 8 · Question 3 part b.iii · 1 mark

2025 JUN · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

3	(a)		• rapid/fast/eq. ; • involuntary/automatic action; • does not involve conscious thought; • protective/prevents (further) damage;	allow instant-mp1 ignore ref to brain ignore sub-conscious ignore unconscious	3
	(b)	(i)	• sensory; • relay/intermediate; • motor;	correct positions on diagram	3
		(ii)	correct direction from sensory to motor;	Mark correctly if there are more than 1 arrow, unle/wrongss they are in opposite direction	1
		(iii)	synapse/synaptic cleft;		1
					Total = 8

Type MS · Show and explain why impulses travel one way across a synapse

Answer 8 of 8 · Question 3 part b.iii · 1 mark

2025 JUN · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks